

Math 110 – Fall 2026

Homework 1

Sets, fields, vector spaces, and subspaces

Suggested completion date: Friday, September 4

Unless otherwise indicated, all vector spaces are over the field $\mathbb{F}$. Problems referring to LADR are from *Linear Algebra Done Right*, 4th ed..

Problem 1. Let $f : X \rightarrow Y$ be a function. For $U \subseteq X$ and $V \subseteq Y$, define

$$f(U) = \{f(x) : x \in U\}, \quad f^{-1}(V) = \{x \in X : f(x) \in V\}.$$

- (a) Prove that $f(f^{-1}(V)) \subseteq V$.
- (b) Prove that $U \subseteq f^{-1}(f(U))$.
- (c) Give examples showing that either inclusion can be strict. Then determine exactly when equality holds in each of (a) and (b).

Problem 2. LADR §1A, Exercise 7.

Problem 3. Let $a, b \in \mathbb{F}$. Prove that if $ab = 0$, then $a = 0$ or $b = 0$.

Problem 4. LADR §1B, Exercise 2: if $a \in \mathbb{F}$ and v is a vector such that $av = 0$, prove that $a = 0$ or $v = 0$.

Problem 5. LADR §1C, Exercise 4.

Problem 6. LADR §1C, Exercise 5.

Problem 7. Work over the field $\mathbb{F}_2 = \mathbb{Z}/2\mathbb{Z}$. Expand $(x + 1)^8$ by first computing $(x + 1)^2$ and then repeatedly squaring. Explain where the calculation differs from the corresponding calculation over $\mathbb{R}$.

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Homework 2

Linear independence, span, bases, and dimension

Suggested completion date: Friday, September 11

Unless otherwise indicated, all vector spaces are over the field $\mathbb{F}$. Problems referring to LADR are from *Linear Algebra Done Right*, 4th ed..

Problem 1. (LADR §2A, Exercise 11.) Prove or give a counterexample: if $v_1, \dots, v_n$ and $w_1, \dots, w_n$ are linearly independent lists in V , then

$$v_1 + w_1, \dots, v_n + w_n$$

is linearly independent.

Problem 2. Let $\mathbb{F}^\infty$ be the vector space of all sequences

$$(a_1, a_2, a_3, \dots), \quad a_j \in \mathbb{F},$$

with coordinatewise addition and scalar multiplication. Let e_j denote the sequence with 1 in the j th coordinate and 0 elsewhere.

- (a) Verify that $\mathbb{F}^\infty$ is a vector space over $\mathbb{F}$.
- (b) Prove that $\mathbb{F}^\infty$ is infinite-dimensional.
- (c) Prove that $e_1, e_2, e_3, \dots$ do *not* span $\mathbb{F}^\infty$.

Problem 3. (LADR §2B, Exercise 5.) Suppose V is finite-dimensional and U, W are subspaces such that $V = U + W$. Prove that V has a basis all of whose vectors belong to $U \cup W$.

Problem 4. Let $\mathbb{F}$ be a field.

- (a) Prove that every nonzero proper subspace of $\mathbb{F}^2$ has the form $\text{span}(v)$ for some nonzero $v \in \mathbb{F}^2$.
- (b) Let $v, w \in \mathbb{F}^2$. Prove that v, w form a basis of $\mathbb{F}^2$ if and only if neither vector is a scalar multiple of the other.

Problem 5. (LADR §2C, Exercise 12.) Suppose U, W are subspaces of $\mathbb{R}^8$ such that

$$\dim U = 3, \quad \dim W = 5, \quad U + W = \mathbb{R}^8.$$

Prove that $\mathbb{R}^8 = U \oplus W$.

Problem 6. Define

$$U = \{p \in \mathcal{P}_4(\mathbb{R}) : p(1) = p'(1) = 0\}.$$

- (a) Prove that U is a subspace of $\mathcal{P}_4(\mathbb{R})$.
- (b) Find a basis of U and determine $\dim U$ directly, without using rank-nullity.

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Homework 3

Linear maps, null spaces, ranges, and matrices

Suggested completion date: Friday, September 18

Unless otherwise indicated, all vector spaces are over the field $\mathbb{F}$. Problems referring to LADR are from *Linear Algebra Done Right*, 4th ed..

Problem 1. Let $T : V \rightarrow W$ be linear, with V finite-dimensional, and let U be a subspace of W .

(a) Prove that $T^{-1}(U) = \{v \in V : Tv \in U\}$ is a subspace of V .

(b) Prove the dimension formula

$$\dim T^{-1}(U) = \dim \text{null } T + \dim(U \cap \text{range } T).$$

(c) Let $D : \mathcal{P}_4(\mathbb{R}) \rightarrow \mathcal{P}_4(\mathbb{R})$ be differentiation, $Dp = p'$. Determine, using part (b), the dimension of the space of polynomials p for which

$$p'(x) = ax^2 + b$$

for some $a, b \in \mathbb{R}$.

Problem 2.

(a) Find a linear map $T : \mathbb{R}^4 \rightarrow \mathbb{R}^4$ such that $\text{null } T = \text{range } T$.

(b) Prove that no linear map $T : \mathbb{R}^5 \rightarrow \mathbb{R}^5$ can satisfy $\text{null } T = \text{range } T$.

Problem 3. Return to the subspace from Homework 2,

$$U = \{p \in \mathcal{P}_4(\mathbb{R}) : p(1) = p'(1) = 0\}.$$

Define a suitable linear map $S : \mathcal{P}_4(\mathbb{R}) \rightarrow \mathbb{R}^2$ having U as its null space, and use rank-nullity to determine $\dim U$. Compare this calculation with the direct basis calculation from Homework 2.

Problem 4. Let $T : V \rightarrow W$ be linear.

(a) Prove that if $v_1, \dots, v_n$ is linearly dependent, then $Tv_1, \dots, Tv_n$ is linearly dependent.

(b) Give an example showing that a linearly independent list need not be sent to a linearly independent list.

Problem 5. Define $T : \mathcal{P}_3(\mathbb{R}) \rightarrow \mathcal{P}_4(\mathbb{R})$ by

$$(Tp)(x) = x^2 p'(x) + p(x).$$

(a) Find the matrix of T with respect to the standard monomial bases

$$1, x, x^2, x^3 \quad \text{and} \quad 1, x, x^2, x^3, x^4.$$

(b) Determine $\text{null } T$ and $\text{range } T$, and give their dimensions.

- (c) Explain directly from your answer to (a) why the range of T is the span of the columns of its representing matrix.

Problem 6. Solve the system

$$\begin{aligned}x + y + z + w &= 1, \\x + z &= 0\end{aligned}$$

over the field $\mathbb{F}_2$. List all solutions.

Problem 7 (optional). Let A be an $n \times m$ matrix over $\mathbb{F}_2$, and suppose $Ax = b$ is consistent. Express the number of solutions in terms of m and $\text{rank } A$.

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Homework 4

Isomorphisms, inverses, products, and quotient spaces

Suggested completion date: Friday, September 25

Unless otherwise indicated, all vector spaces are over the field $\mathbb{F}$. Problems referring to LADR are from *Linear Algebra Done Right*, 4th ed..

Problem 1. Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

(a) Verify by direct multiplication that

$$A \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = (ad - bc)I.$$

- (b) Define $\det A = ad - bc$. Prove that if $\det A \neq 0$, then A is invertible and derive the usual formula for A^{-1} .
- (c) Prove conversely, without using any general determinant theory, that if $\det A = 0$, then A is not invertible.

Problem 2. Use the result of Problem 1 to decide which of the following matrices are invertible, and find the inverse whenever it exists:

$$\begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}, \quad \begin{pmatrix} 1 & i \\ i & -1 \end{pmatrix} \text{ over } \mathbb{C}, \quad \begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix} \text{ over } \mathbb{F}_5.$$

Problem 3. (LADR §3D, Exercise 3.) Let $T : V \rightarrow V$ be linear and suppose V is finite-dimensional. Prove that the following are equivalent:

- (a) T is an isomorphism;
- (b) for every basis $v_1, \dots, v_n$ of V , the list $Tv_1, \dots, Tv_n$ is a basis of V .

Problem 4. Suppose B and C are matrices for which BC is defined.

(a) Prove

$$\text{rank}(BC) \leq \min\{\text{rank } B, \text{rank } C\}.$$

(b) Give an example in which the inequality is strict even though both B and C have positive rank.

Problem 5. Let V and W be finite-dimensional vector spaces. Prove

$$\dim(V \times W) = \dim V + \dim W$$

by constructing a basis of $V \times W$ from bases of V and W .

Problem 6. Let U be a subspace of V and let $v, v' \in V$.

- (a) Prove that the affine set $v + U$ is a subspace of V if and only if $v \in U$; in that case $v + U = U$.
- (b) Prove that $v + U = v' + U$ if and only if $v - v' \in U$.

Problem 7 (optional). Suppose U is a subspace of a finite-dimensional vector space V , and $S : U \rightarrow V$ is injective. Prove that S can be extended to an invertible operator $T : V \rightarrow V$.

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Homework 5

Quotient spaces, isomorphism theorems, and duality

Suggested completion date: Friday, October 2

Unless otherwise indicated, all vector spaces are over the field $\mathbb{F}$. Problems referring to LADR are from *Linear Algebra Done Right*, 4th ed..

Problem 1. Define

$$T : \mathcal{P}_3(\mathbb{R}) \longrightarrow \mathbb{R}^2, \quad T(p) = (p(0), p(1)).$$

(a) Determine $\text{null } T$ and prove that T is surjective.

(b) Describe explicitly the induced isomorphism

$$\mathcal{P}_3(\mathbb{R}) / \text{null } T \longrightarrow \mathbb{R}^2$$

supplied by the first isomorphism theorem.

(c) Find a basis of $\mathcal{P}_3(\mathbb{R}) / \text{null } T$.

Problem 2. Let $T : V \rightarrow W$ be linear and let U be a subspace of V contained in $\text{null } T$. Prove that there is a unique linear map

$$\tilde{T} : V/U \longrightarrow W$$

such that $\tilde{T}(v + U) = Tv$ for every $v \in V$. Determine $\text{null } \tilde{T}$ in terms of $\text{null } T$ and U .

Problem 3. (LADR §3F, Exercise 1.) Prove that every linear functional $\varphi : V \rightarrow \mathbb{F}$ is either the zero map or is surjective.

Problem 4. Let $v_1, \dots, v_n$ be a basis of a finite-dimensional vector space V , and let $\varphi_1, \dots, \varphi_n$ be its dual basis.

(a) Prove that every $v \in V$ satisfies

$$v = \sum_{j=1}^n \varphi_j(v) v_j.$$

(b) If $\psi \in V'$, prove that

$$\psi = \sum_{j=1}^n \psi(v_j) \varphi_j.$$

Problem 5. Let $T : V \rightarrow W$ be linear between finite-dimensional spaces. If A is the matrix of T with respect to bases $\mathcal{B}$ of V and $\mathcal{C}$ of W , prove that the matrix of the dual map

$$T' : W' \longrightarrow V', \quad T'(\psi) = \psi \circ T,$$

with respect to the corresponding dual bases is A^T .

Problem 6. Let

$$U = \text{span}((1, 1, 0), (0, 1, 1)) \subseteq \mathbb{R}^3.$$

(a) Determine the annihilator $U^0 \subseteq (\mathbb{R}^3)'$, identifying a functional with its row vector.

(b) Verify in this example that $\dim U + \dim U^0 = 3$.

Problem 7 (optional). Let $j_V : V \rightarrow V''$ be defined by $j_V(v)(\varphi) = \varphi(v)$. Prove that j_V is injective, and prove that it is an isomorphism when V is finite-dimensional.

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Homework 6

Invariant subspaces and eigenvalues

Suggested completion date: Friday, October 9

Unless otherwise indicated, all vector spaces are over the field $\mathbb{F}$. Problems referring to LADR are from *Linear Algebra Done Right*, 4th ed..

Problem 1. Let $P : V \rightarrow V$ satisfy $P^2 = P$.

- (a) Prove that every eigenvalue of P belongs to $\{0, 1\}$.
- (b) Prove that $V = \text{null } P \oplus \text{range } P$.

Problem 2. Let $D : \mathcal{P}_n(\mathbb{F}) \rightarrow \mathcal{P}_n(\mathbb{F})$ be differentiation, where $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$.

- (a) Determine all eigenvalues and eigenspaces of D .
- (b) Explain why the answer is different if one replaces $\mathcal{P}_n$ by the space of smooth functions on $\mathbb{R}$.

Problem 3. Let $S : V \rightarrow V$ be invertible and set $R = S^{-1}TS$. Prove that T and R have the same eigenvalues. More precisely, prove that

$$E(\lambda, R) = S^{-1}E(\lambda, T)$$

for every scalar λ .

Problem 4. Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Using only the 2×2 determinant formula from Homework 4, prove that

$$\det(tI - A) = t^2 - (a + d)t + (ad - bc).$$

Then prove that a scalar λ is an eigenvalue of A if and only if it is a root of this polynomial.

Problem 5. For each of the following matrices, find the eigenvalues and eigenspaces over the indicated field, and decide whether the matrix has a basis of eigenvectors:

$$A_1 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad A_3 = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

For A_1 , work first over $\mathbb{R}$ and then over $\mathbb{C}$; for A_2 and A_3 , work over $\mathbb{R}$.

Problem 6 (optional). Let $T : V \rightarrow V$ and let $v \neq 0$. Define the cyclic subspace

$$\mathcal{C}(v, T) = \text{span}(v, Tv, T^2v, \dots).$$

- (a) Prove that $\mathcal{C}(v, T)$ is T -invariant.
- (b) If V is finite-dimensional, prove that there is some $m \leq \dim V$ such that

$$\mathcal{C}(v, T) = \text{span}(v, Tv, \dots, T^{m-1}v).$$

Problem 7 (optional). Consider the column-stochastic matrix

$$P = \begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1/2 & 1/2 \end{pmatrix}.$$

Find a probability vector p satisfying $Pp = p$, and explain its interpretation for the corresponding three-state Markov chain.

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Homework 7

Diagonalizability and the minimal polynomial

Suggested completion date: Friday, October 16

Unless otherwise indicated, all vector spaces are over the field $\mathbb{F}$. Problems referring to LADR are from *Linear Algebra Done Right*, 4th ed..

Problem 1. Return to the matrices from Homework 6

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

- (a) Find the minimal polynomial of A over $\mathbb{R}$ and over $\mathbb{C}$.
- (b) Find the minimal polynomial of B over $\mathbb{R}$.
- (c) Use the minimal-polynomial criterion for diagonalizability to recover the conclusions of Homework 6 about whether these operators are diagonalizable over the indicated fields.

Problem 2. Let $D : \mathcal{P}_3(\mathbb{R}) \rightarrow \mathcal{P}_3(\mathbb{R})$ be differentiation.

- (a) Find the minimal polynomial of D .
- (b) Decide whether D is diagonalizable.

Problem 3. Let T be an invertible operator on a finite-dimensional vector space, and let

$$m_T(x) = x^d + a_{d-1}x^{d-1} + \cdots + a_1x + a_0$$

be its minimal polynomial.

- (a) Prove that $a_0 \neq 0$.
- (b) Prove that T^{-1} is a polynomial in T .
- (c) Suppose now that V is a complex vector space and that

$$m_T(x) = x^4 + x^3 - x^2 - 5x + 4.$$

Express T^{-1} explicitly as a polynomial in T , find the eigenvalues of T , and decide whether T is diagonalizable.

Problem 4 (optional). Let m_T be the minimal polynomial of an operator T .

- (a) Prove that the minimal polynomial of $T - \lambda I$ is $m_T(x + \lambda)$.
- (b) If $c \neq 0$ and $d = \deg m_T$, prove that the minimal polynomial of cT is

$$c^d m_T(x/c).$$

Problem 5. Let A be a 2×2 complex matrix whose only eigenvalue is λ . Prove that its minimal polynomial is either

$$x - \lambda \quad \text{or} \quad (x - \lambda)^2.$$

Show that the first case occurs exactly when $A = \lambda I$, and conclude that A is diagonalizable if and only if $A = \lambda I$.

Problem 6. Consider

$$C = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

Find its eigenvalues, eigenspaces, and minimal polynomial. Decide whether C is diagonalizable.

Problem 7 (optional). Let T be an operator whose minimal polynomial splits as

$$m_T(x) = (x - \lambda_1) \cdots (x - \lambda_r)$$

with distinct roots. For a fixed j , construct a polynomial p_j such that $p_j(\lambda_j) = 1$ and $p_j(\lambda_k) = 0$ for $k \neq j$, and explain why $p_j(T)$ is the projection onto $E(\lambda_j, T)$ along the sum of the other eigenspaces.

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Homework 8

Generalized eigenvectors and Jordan form

Suggested completion date: Friday, October 23

Unless otherwise indicated, all vector spaces are over the field $\mathbb{F}$. Problems referring to LADR are from *Linear Algebra Done Right*, 4th ed..

Problem 1. Let $D : \mathcal{P}_n(\mathbb{C}) \rightarrow \mathcal{P}_n(\mathbb{C})$ be differentiation.

- (a) Prove that D is nilpotent and determine its nilpotency index.
- (b) Determine the geometric and algebraic multiplicities of the eigenvalue 0.
- (c) Find a basis with respect to which the matrix of D is a single Jordan block.

Problem 2. Construct explicit matrices having the following spectral data.

- (a) Eigenvalue 2 has algebraic multiplicity 3 and geometric multiplicity 1; eigenvalue -1 has algebraic multiplicity 2 and geometric multiplicity 2.
- (b) The same algebraic multiplicities as in (a), but the geometric multiplicity of 2 is 2.

For each matrix, give its minimal polynomial.

Problem 3. Let

$$J = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

- (a) Compute $\text{null}(J - I)$, $\text{null}((J - I)^2)$, and $\text{null}((J - I)^3)$.
- (b) Determine the generalized eigenspace $G(1, J)$ and the geometric and algebraic multiplicities of 1.
- (c) Compute J^m for every positive integer m .

Problem 4. Let N be nilpotent with Jordan form consisting of one block of size 3 and one block of size 2.

- (a) Compute $\dim \text{null } N^k$ for $k = 0, 1, 2, 3, 4$.
- (b) Suppose instead that you are only told

$$\dim \text{null } N = 2, \quad \dim \text{null } N^2 = 4, \quad \dim \text{null } N^3 = 5.$$

Explain how these numbers force the same two block sizes.

Problem 5 (optional). Let $T \in \mathcal{L}(\mathbb{C}^6)$ have eigenvalues 1 and 2, and suppose

$$\dim \text{null}(T - I) = 1, \quad \dim \text{null}((T - I)^2) = 2, \quad \dim \text{null}((T - I)^3) = 2,$$

while

$$\begin{aligned}\dim \text{null}(T - 2I) &= 2, & \dim \text{null}((T - 2I)^2) &= 3, \\ \dim \text{null}((T - 2I)^3) &= 4, & \dim \text{null}((T - 2I)^4) &= 4.\end{aligned}$$

Determine the Jordan normal form of T , up to reordering of the blocks.

Problem 6 (optional). Suppose an operator has Jordan blocks

$$J_3(4), \quad J_2(4), \quad J_1(-1).$$

Determine:

- (a) its characteristic polynomial;
- (b) its minimal polynomial;
- (c) the geometric and algebraic multiplicities of each eigenvalue;
- (d) whether it is diagonalizable.

Problem 7 (optional). Prove that for a nilpotent operator N ,

$$\dim \text{null } N^k - \dim \text{null } N^{k-1}$$

is the number of Jordan blocks of size at least k .

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Homework 9

Trace and inner products

Suggested completion date: Friday, October 30

Unless otherwise indicated, all vector spaces are over the field $\mathbb{F}$. Problems referring to LADR are from *Linear Algebra Done Right*, 4th ed..

Problem 1. Let P be a projection on a finite-dimensional vector space, so that $P^2 = P$. Prove that

$$\operatorname{tr} P = \operatorname{rank} P.$$

You may choose a basis adapted to the decomposition $V = \operatorname{null} P \oplus \operatorname{range} P$.

Problem 2. Let $T \in \mathcal{L}(\mathbb{C}^5)$ have minimal polynomial

$$m_T(x) = (x - 2)^2(x + 1).$$

- (a) Determine all possible algebraic multiplicities of the eigenvalues 2 and -1 .
- (b) Determine all possible values of $\operatorname{tr} T$.
- (c) For each possible trace, exhibit a Jordan normal form realizing it.

Problem 3.

- (a) Prove that every nilpotent operator on a finite-dimensional complex vector space has trace 0.
- (b) Give a nonzero operator with trace 0 that is not nilpotent.
- (c) Give two 2×2 matrices A, B for which $AB - BA \neq 0$, and verify directly that $\operatorname{tr}(AB - BA) = 0$.

Problem 4. For each of the following formulas on $\mathbb{R}^2$, decide whether it defines an inner product. Justify your answer.

$$\langle x, y \rangle_1 = x_1 y_1 + 2x_2 y_2,$$

$$\langle x, y \rangle_2 = x_1 y_1 - x_2 y_2,$$

$$\langle x, y \rangle_3 = (x_1 + x_2)(y_1 + y_2).$$

Problem 5. On $\mathcal{P}_2(\mathbb{R})$ define

$$\langle p, q \rangle = \int_{-1}^1 p(x)q(x) dx.$$

- (a) Prove that this is an inner product.
- (b) Compute the norms of 1, x , and x^2 .
- (c) Determine which pairs among 1, x , x^2 are orthogonal.

Problem 6. Use the Cauchy–Schwarz inequality to prove that for all real numbers a, b, c ,

$$(a + b + c)^2 \leq 3(a^2 + b^2 + c^2).$$

Determine exactly when equality holds.

Problem 7 (optional). Prove the parallelogram identity

$$\|u + v\|^2 + \|u - v\|^2 = 2\|u\|^2 + 2\|v\|^2$$

in every real or complex inner-product space.

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Homework 10

Orthogonality, Gram–Schmidt, and projections

Suggested completion date: Friday, November 6

Unless otherwise indicated, all vector spaces are over the field $\mathbb{F}$. Problems referring to LADR are from *Linear Algebra Done Right*, 4th ed..

Problem 1. On $\mathcal{P}_2(\mathbb{R})$ use the inner product

$$\langle p, q \rangle = \int_{-1}^1 p(x)q(x) dx.$$

Apply the Gram–Schmidt procedure to the ordered basis

$$1, x, x^2$$

to obtain an orthonormal basis of $\mathcal{P}_2(\mathbb{R})$.

Problem 2. Let

$$W = \text{span}((1, -1, 1, 0), (0, 2, 1, 0)) \subseteq \mathbb{R}^4$$

with the standard inner product. Find a basis of $W^\perp$.

Problem 3. On $\mathcal{P}_2(\mathbb{R})$ use the same inner product as in Problem 1, and let

$$W = \text{span}(x + 1).$$

Find a basis of $W^\perp$.

Problem 4. Let U and W be subspaces of a finite-dimensional inner-product space V . Prove

$$(U \cap W)^\perp = U^\perp + W^\perp.$$

Problem 5. Let U be a subspace of a finite-dimensional inner-product space V , and let $e_1, \dots, e_m$ be an orthonormal basis of U .

(a) Prove that

$$P_U v = \sum_{j=1}^m \langle v, e_j \rangle e_j.$$

(b) Prove the best-approximation property

$$\|v - P_U v\| \leq \|v - u\| \quad \text{for every } u \in U,$$

and determine when equality holds.

Problem 6 (optional). On $\mathcal{P}_1(\mathbb{R})$ use the inner product

$$\langle f, g \rangle = \int_{-1}^1 f(x)g(x) dx.$$

Find the unique polynomial $p \in \mathcal{P}_1(\mathbb{R})$ such that

$$\int_0^2 f(x) dx = \langle f, p \rangle$$

for every $f \in \mathcal{P}_1(\mathbb{R})$.

Problem 7 (optional). Let P be a projector on a finite-dimensional inner-product space, so $P^2 = P$. Prove that P is an orthogonal projection if and only if

$$\|Pv\| \leq \|v\| \quad \text{for every } v.$$

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Homework 11

Adjoints, self-adjoint and normal operators, and the spectral theorem

Suggested completion date: Friday, November 13

Unless otherwise indicated, all vector spaces are over the field $\mathbb{F}$. Problems referring to LADR are from *Linear Algebra Done Right*, 4th ed..

Problem 1. Let $T : \mathbb{F}^n \rightarrow \mathbb{F}^n$ be defined by

$$T(z_1, \dots, z_n) = (0, z_1, \dots, z_{n-1}),$$

where $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$. Find T^* .

Problem 2. Let $T : V \rightarrow W$ be a linear map between finite-dimensional inner-product spaces. Prove that T is injective if and only if T^* is surjective.

Problem 3. Let $T : \mathbb{F}^2 \rightarrow \mathbb{F}^2$ be the quarter-turn

$$T(x, y) = (-y, x).$$

- (a) Find T^* .
- (b) Prove that T is normal but not self-adjoint.
- (c) Find all eigenvalues of T when $\mathbb{F} = \mathbb{R}$, and when $\mathbb{F} = \mathbb{C}$.

Problem 4. Let P be a projector on a finite-dimensional inner-product space. Prove that the following are equivalent:

- (a) P is self-adjoint;
- (b) P is normal;
- (c) P is the orthogonal projection onto range P .

Problem 5. Let T be self-adjoint and let U be a T -invariant subspace. Prove that $U^\perp$ is also T -invariant.

Problem 6. Consider

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

- (a) Find an orthonormal basis of $\mathbb{R}^2$ consisting of eigenvectors of A .
- (b) Find an orthogonal matrix Q and diagonal matrix D such that

$$A = QDQ^T.$$

- (c) Use this diagonalization to find a formula for A^m for every positive integer m .

Problem 7 (optional). Let T be a normal operator on a finite-dimensional complex inner-product space and let p be a polynomial. Use the complex spectral theorem to prove that $p(T)$ is normal.

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Homework 12

Positive operators, isometries, polar decomposition, and singular values

Suggested completion date: Friday, November 20

Unless otherwise indicated, all vector spaces are over the field $\mathbb{F}$. Problems referring to LADR are from *Linear Algebra Done Right*, 4th ed..

Problem 1. Classify each of the following real symmetric matrices as positive definite, positive semidefinite but not positive definite, or not positive semidefinite:

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Problem 2. Find the positive square root of

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

Your answer should be an explicit 2×2 matrix.

Problem 3. Let $S : V \rightarrow W$ be linear between finite-dimensional inner-product spaces, and let $e_1, \dots, e_n$ be an orthonormal basis of V . Prove that S is an isometry if and only if

$$Se_1, \dots, Se_n$$

is an orthonormal list in W .

Problem 4. Find the polar decomposition

$$A = S\sqrt{A^*A}$$

of

$$A = \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$$

acting on $\mathbb{R}^2$ with its standard inner product.

Problem 5. Find a singular value decomposition of

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Give the singular values and orthonormal bases e_1, e_2 and f_1, f_2 such that

$$Ae_j = s_j f_j, \quad j = 1, 2.$$

Problem 6 (optional). Let T be an invertible operator on a finite-dimensional inner-product space. Prove that the singular values of T^{-1} are the reciprocals of the singular values of T .

Problem 7 (optional). Let T have singular values $s_1 \geq \dots \geq s_n \geq 0$. Prove that

$$\|Tv\| \leq s_1\|v\|$$

for every v , and show that equality occurs for some nonzero v .

Math 110 – Fall 2026

Homework 13

Complexification, real normal operators, and determinant

Suggested completion date: Friday, December 4

Unless otherwise indicated, all vector spaces are over the field $\mathbb{F}$. Problems referring to LADR are from *Linear Algebra Done Right*, 4th ed..

Problem 1. Let V be a finite-dimensional real vector space with basis $v_1, \dots, v_n$.

- (a) Prove that the same list is a basis of the complexification $V_{\mathbb{C}}$ over $\mathbb{C}$.
- (b) If $T \in \mathcal{L}(V)$, prove that the matrix of $T_{\mathbb{C}}$ with respect to this basis is the same matrix as that of T .

Problem 2. Let $T \in \mathcal{L}(V)$ for a finite-dimensional real vector space V . Prove that T and its complexification $T_{\mathbb{C}}$ have the same minimal polynomial.

Problem 3. Let

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

act on $\mathbb{R}^2$, and consider its complexification.

- (a) Find the two complex eigenvalues and corresponding eigenvectors of $J_{\mathbb{C}}$.
- (b) Explain how the two eigenvectors are related by complex conjugation.
- (c) Explain how this example gives the real 2×2 rotation block associated with a conjugate pair of complex eigenvalues.

Problem 4. Let

$$A = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$$

act on $\mathbb{R}^2$.

- (a) Prove directly that A is normal.
- (b) Find the eigenvalues of $A_{\mathbb{C}}$.
- (c) Prove that A is an isometry if and only if $a^2 + b^2 = 1$.

Problem 5. Let T be an operator on a two-dimensional vector space with basis e_1, e_2 , and suppose

$$\mathcal{M}(T) = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Using the alternating-form definition of determinant, recover the formula

$$\det T = ad - bc.$$

Problem 6 (optional). Let T be an operator on a finite-dimensional complex vector space. Using a Jordan basis and the formula for the determinant of an upper-triangular matrix, prove

$$\det(zI - T) = \chi_T(z),$$

where χ_T is the characteristic polynomial defined earlier using algebraic multiplicities.

Problem 7 (optional). Let

$$F_0 = 0, \quad F_1 = 1, \quad F_{n+1} = F_n + F_{n-1},$$

and let

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

Using

$$A^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix} \quad (n \geq 1)$$

and multiplicativity of determinant, prove Cassini's identity

$$F_{n-1}F_{n+1} - F_n^2 = (-1)^n.$$

Problem 8 (optional). Let T have singular values $s_1, \dots, s_n$. Using the singular value decomposition, prove

$$|\det T| = s_1 \cdots s_n.$$